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Velo3D and the New Arsenal of American Manufacturing

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Velo3D and the New Arsenal of American Manufacturing

Why the Real Defense Bottleneck Is Not Demand for Weapons — But the Ability to Make the Right Parts, in the Right Place, at the Right Time

The easiest way to misunderstand Velo3D is to call it a 3D-printing company.

That description is technically true.

It is also strategically too small.

The more useful way to think about Velo3D in 2026 is this: **it is trying to become part of the production infrastructure that allows the United States to rebuild a more resilient defense and aerospace supply chain.**

Its value does not come mainly from selling machines.

It comes from the possibility that it becomes **one of the few trusted ways to manufacture complex metal parts quickly, domestically, and repeatedly** for programs where delay is expensive and failure is unacceptable.

Velo3D's own 2025 results and 2026 outlook are now centered on exactly that shift, with management emphasizing defense and aerospace adoption, backlog growth, and Rapid Production Solutions (RPS) as the engine of its next phase.

Our view is straightforward:

The Velo3D story is no longer best framed as "additive manufacturing eventually changes industry." That thesis has been around for years and has often disappointed.

The better thesis is narrower and more investable:

> **America is rediscovering that manufacturing bottlenecks are a national-security problem, and Velo3D is trying to position itself where that pain is most urgent.**

If that positioning proves durable, the stock could be re-rated as a strategic manufacturing platform rather than a speculative equipment story.

If it does not, it remains what many investors still fear it is: **an interesting technology company with fragile economics.**

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1. The Real Problem in Defense Manufacturing Is Not "Capacity" in the Abstract

A lot of investors hear "defense buildout" and instinctively think in terms of larger weapons budgets — more missiles, more aircraft, more ships.

That is true at the top line, but it misses the harder issue underneath.

The United States does not only need more systems.

It needs **the ability to manufacture and replace small, complex, often low-volume parts that legacy suppliers may no longer make economically or quickly.**

That is the logic behind Velo3D's recent defense narrative.

In its March 2026 results release, the company said it secured a $32.6 million contract supporting the Defense Innovation Unit's **Project FORGE**, which is aimed at solving manufacturing bottlenecks for traditionally manufactured parts and platforms inside the U.S. defense industrial base.

The company also highlighted an $11.5 million multi-year full-rate production RPS contract from a key U.S. defense prime contractor and a qualification milestone with the U.S. Army's Ground Vehicle Systems Center (GVSC).

These are not theoretical "innovation lab" signals.

They suggest the company is being pulled into **the production and sustainment side of the defense stack.**

That distinction matters because the highest-value defense suppliers are rarely the ones with the best slide decks.

They are the ones that get embedded into the parts of the system where time-to-production and repeatability matter more than novelty.

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2. Velo3D's Moat Is Partly Technical — But Increasingly Geopolitical

There is a temptation to evaluate additive manufacturing companies mainly on print specs, design complexity, material capability, or machine throughput.

Those things matter, but in Velo3D's case they are no longer the whole story.

The bigger part of the moat may be **geopolitical and institutional.**

In February 2026, Velo3D announced it had become the first additive manufacturing vendor qualified to support the U.S. Army GVSC qualification initiative. The company said it met all qualification criteria in less than two weeks, and the program covers critical components printed on Velo3D's Sapphire platforms in both Aluminum CP1 and Inconel 718.

Upon successful completion, the qualified AM alternatives can be inserted into Army supply chains through TACOM.

That is not just a technical validation.

It is a **trust signal.**

For defense and aerospace, trust signals matter as much as machine capability.

The government is not simply buying a cheaper part. It is buying a manufacturing method it believes can connect to mission-critical supply chains without creating new vulnerabilities.

Velo3D's own investor materials now emphasize that it can securely connect to military networks and support qualification efforts in a way that aligns with U.S. defense requirements.

That makes the company more than a machine vendor.

It makes it a potential participant in **the digital re-architecture of defense manufacturing.**

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3. The Business-Model Shift Matters More Than the Hardware Story

The most important change in Velo3D may not be the printer.

It may be the business model.

Historically, additive-manufacturing names have often struggled because:

• Equipment sales are lumpy

• Customers move slowly

• The gap between technical promise and scaled utilization is wide

Velo3D is trying to narrow that gap through **Rapid Production Solutions (RPS)** — a model where it does not simply sell a machine and wait for the customer to use it effectively. Instead, it manufactures the part directly and becomes part of the customer's production pathway."

In its Q2 2025 results, the company said RPS bookings surged 79% quarter over quarter, and in its 2025 year-end release management said strong demand for RPS drove double-digit growth and will represent an increasing share of revenue going forward.

This is strategically important.

• Selling printers is **a capital-equipment business**

• Selling production output is **a supply-chain business**

Those are very different valuation frameworks.

If RPS continues to scale, Velo3D could migrate from being viewed as a cyclical machine supplier to being viewed as **a specialized domestic manufacturing capacity provider.**

That would make its revenue stream more directly tied to actual defense and aerospace throughput rather than just customer capital budgets.

Management now says system sales should still remain the primary revenue driver in 2026, but it also says RPS will contribute an increasing share of revenue under the new go-to-market strategy.

That is the right direction, even if it is not yet fully reflected in the numbers.

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4. "Adoption" Is Not the Key Word Anymore. "Dependence" Is.

A lot of technology companies get rewarded too early for "adoption."

Adoption can mean pilots, prototypes, one-off validation jobs, or press releases that never turn into recurring demand.

The more meaningful word is **dependence.**

That is where the Velo3D case becomes interesting.

The company's recent defense announcements suggest some programs are moving beyond experimentation into production support. Its March 2026 release states that certain programs can scale from a single production system to multiple systems within months, and that the compounding effect on capacity requirements is significant.

This is not yet proof of a durable moat.

But it is the first stage of something investors should take seriously: **customers beginning to structure real throughput assumptions around Velo3D's capacity.**

Once a defense or aerospace program begins to rely on a supplier not just for design flexibility but for actual production continuity, the relationship changes.

The question is no longer, "Can this technology work?"

It becomes:

> **"Can this supplier keep the line moving?"**

That is a much more valuable place to sit in a supply chain.

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5. Project FORGE Is Bigger Than One Contract

Investors should not read Project FORGE as just another government contract award.

They should read it as **evidence of where the Pentagon thinks the real pain is.**

The official company description says Project FORGE is charged with identifying solutions to manufacturing bottlenecks for traditionally manufactured parts and platforms in the U.S. defense industrial base.

That tells you something important:

The problem is not simply one of engineering.

It is one of **throughput, localization, and responsiveness.**

A system can be technologically advanced and still strategically fragile if the part replacement cycle is too slow.

That is why additive manufacturing is becoming more strategically relevant than many investors assumed a few years ago.

In a peacetime mindset, older methods can appear "good enough."

In a contested supply environment, the premium shifts toward methods that:

• Shorten lead times

• Reduce dependence on fragile vendor networks

• Allow domestic production to restart around digital files rather than around long-lost tooling

This is especially relevant for sustainment and MRO. Decades-old tanks, aircraft, or naval systems may require rapid reproduction of difficult parts. That is one of the most compelling practical use cases for defense AM.

If Velo3D becomes one of the trusted pathways for that, the market may eventually stop classifying it as a niche additive company and start classifying it as **part of the digital arsenal of U.S. manufacturing resilience.**

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6. The Company Is Still Financially Fragile — And That Matters

Now the hard part.

Velo3D's strategic positioning has improved meaningfully.

Its financial profile is still weak.

The company reported:

• $46.0 million of full-year 2025 revenue

• $31 million backlog

• Cash of $39.0 million at year-end after a private placement and debt-to-equity conversions

Guidance for 2026:

• $60 million to $70 million of revenue

• EBITDA positive in the second half of 2026 (delayed from first half)

Gross margin for full-year 2025 was still **negative 16.1%**, and the company recorded a $71.4 million GAAP net loss. It also disclosed that it expects to raise additional capital to support capacity expansion.

Those facts are not side issues.

They are central to the equity case.

This means investors should be careful not to confuse strategic importance with financial maturity. Velo3D may be moving into more valuable parts of the defense and aerospace ecosystem, but **it has not yet proven that this can be translated into a high-quality, self-funding operating model.**

The 2026 thesis rests heavily on execution:

• Higher RPS mix

• Better pricing

• Less overhead drag

• Better capacity utilization

• Margin improvement above 30% in the second half of the year

That is plausible.

It is not yet bankable.

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7. The Real Investment Question

The question is not "Does 3D printing matter?" — because the answer is already yes in some contexts.

The more valuable question is:

> **Which additive manufacturing firms become indispensable in the places where repeatability, material qualification, security, and domestic production matter most?**

This is where Velo3D stands apart from the generic AM bucket.

It is not trying to win a broad consumer or industrial "3D printing revolution."

It is trying to win a narrower, more valuable role inside **mission-critical metal manufacturing** where defense and aerospace customers increasingly care about localized production and trusted qualification.

That is a better market than the one many additive names chased in the past.

It is smaller, but more defensible.

The American analogy here is not a flashy Silicon Valley disruptor.

It is something closer to **a new industrial subcontractor for the digital arsenal** — a company that becomes essential not because it is famous, but because the machinery of national production starts routing through it.

That can be a very powerful position if it works.

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8. What U.S. Investors Should Watch

For investors, there are four things that matter from here.

First, whether RPS actually becomes a larger share of revenue in 2026 and whether that translates into structurally better gross margins. If the business model changes but margins do not, the re-rating case weakens.

Second, whether defense and aerospace orders convert from high-profile announcements into repeatable production cadence. Velo3D's recent contract wins are impressive, but the market will eventually demand evidence of execution at scale.

Third, whether the company can raise capital on terms that preserve enough upside for shareholders. The need for more financing is clearly disclosed, and dilution remains a live risk.

Fourth, whether Velo3D continues to deepen its institutional trust with the Army, primes, and aerospace customers. In a strategic manufacturing theme, trust can become a bigger moat than technology alone.

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9. Investment Conclusion

The strongest version of the Velo3D bull case is not that 3D printing finally gets its moment.

It is that the United States is rebuilding parts of its defense and aerospace supply chain around domestic, qualified, digitally driven production — and Velo3D is trying to become one of the core manufacturing nodes inside that rebuild.

That makes the company **more strategically important than its market capitalization or legacy category labels suggest.**

But it also leaves no room for romanticism.

This is still a high-risk equity. It needs to prove margin recovery, operational execution, and financing discipline. Strategic relevance alone will not save shareholders if the economics do not improve.

Our own view is constructive but conditional.

The company has moved into a much more investable narrative than it had when it was just another additive-manufacturing story.

The right way to frame it now is this:

> **Velo3D is not primarily a 3D-printing company anymore. It is a candidate infrastructure layer for America's resilient defense manufacturing base.**

If that transition holds up in the numbers, the market may eventually treat it very differently.

Final Line

The real bet on Velo3D is not that additive manufacturing becomes fashionable.

It is that **America decides speed, domestic control, and repeatable production are now worth paying up for** — and Velo3D becomes one of the few companies already standing in that lane.

This Deep Dive is part of Brutal Edge's "American Infrastructure Revival" framework series. Related analysis: Intel's Great Repricing (semiconductor manufacturing), JPMorgan American Fortress (financial infrastructure).